

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTMES304 - MATERIALS SCIENCE AND METALLURGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Following metal(s) possess(es) HCP crystal structure: (a) Zinc (b) Titanium (c) Cobalt (d) All of these
- 2 Cracks are ----- defects. (a) 0-D (b) 1-D (c) 2-D (d) 3-D
- 3 The number of favorable slip systems is highest in ----- crystal structures. (a) BCC (b) FCC (c) HCP (d) None of these
- 4 ----- is the ability of the material to absorb shocks. (a) Toughness (b) Stiffness (c) Hardness (d) None of these
- 5 Tension test is used for evaluating the properties of ----- metals. (a) Ductile (b) Brittle (c) Both 'a' and 'b' (d) None of these
- 6 ----- hardness test gives more accurate readings. (a) Brinell (b) Rockwell (c) Both 'a' and 'b' (d) None of these
- 7 Lower critical temperature for steel is ----- degree Celsius. (a) 210 (b) 727 (c) 910 (d) 1147
- 8 Following test is used for the evaluation of hardenability: (a) Brinell (b) Vickers (c) Rockwell (d) Jominy
- 9 Normalizing process involves cooling of the object in ----- . (a) air (b) oil (c) water (d) furnace
- 10 Following surface hardening treatment is used for depositing both C and N into the components' surface: (a) Carburizing (b) Nitriding (c) Carbonitriding (d) None of these
- 11 Dye penetrant test is useful for detection of flaws in ----- materials. (a) Ferrous (b) Nonferrous (c) Ceramics (d) All of these
- 12 Following is/are the strengthening mechanism(s): (a) Refinement of grain size (b) Solid solution strengthening (c) Cold working (d) All of these

Q2. Solve the following.

- A) What are the three basic types of crystal structures? Explain any two of those w.r.t. schematic diagram, and atomic packing factor. **6**
- B) Classify the crystal defects and explain any two using schematic diagrams. **6**

Q3. Solve the following.

- A) Compare the Brinell hardness test with Rockwell hardness test with the help of schematic diagrams. 6
- B) Draw a neat diagram for Fe-Fe₃C equilibrium diagram. Show all the phases, transformations and important temperatures on it. 6

Q4. Solve Any Two of the following.

- A) Classify steels and explain any two types in detail. 6
- B) Discuss the following heat treatment processes in detail: (a) Annealing (b) Normalizing (c) Hardening. 6
- C) Describe the following surface hardening processes in detail: (a) Carburizing (b) Nitriding (c) Induction hardening. 6

Q5. Solve Any Two of the following.

- A) Explain the procedure of specimen preparation for microscopy. Compare macroscopy with microscopy. 6
- B) Explain principle and working of optical metallurgical microscope. 6
- C) Describe the spark test w.r.t. purpose, mechanism of generation of spark, and the spark patterns for low carbon and high carbon steel. 6

Q6. Solve Any Two of the following.

- A) What are the various strengthening mechanisms? Explain any two in detail. 6
- B) Explain dye penetrant test with the help of schematic diagrams. 6
- C) Describe magnetic particle inspection and ultrasonic test in brief. 6

*** End ***